

Special sense-Physiology
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Taste and Smell

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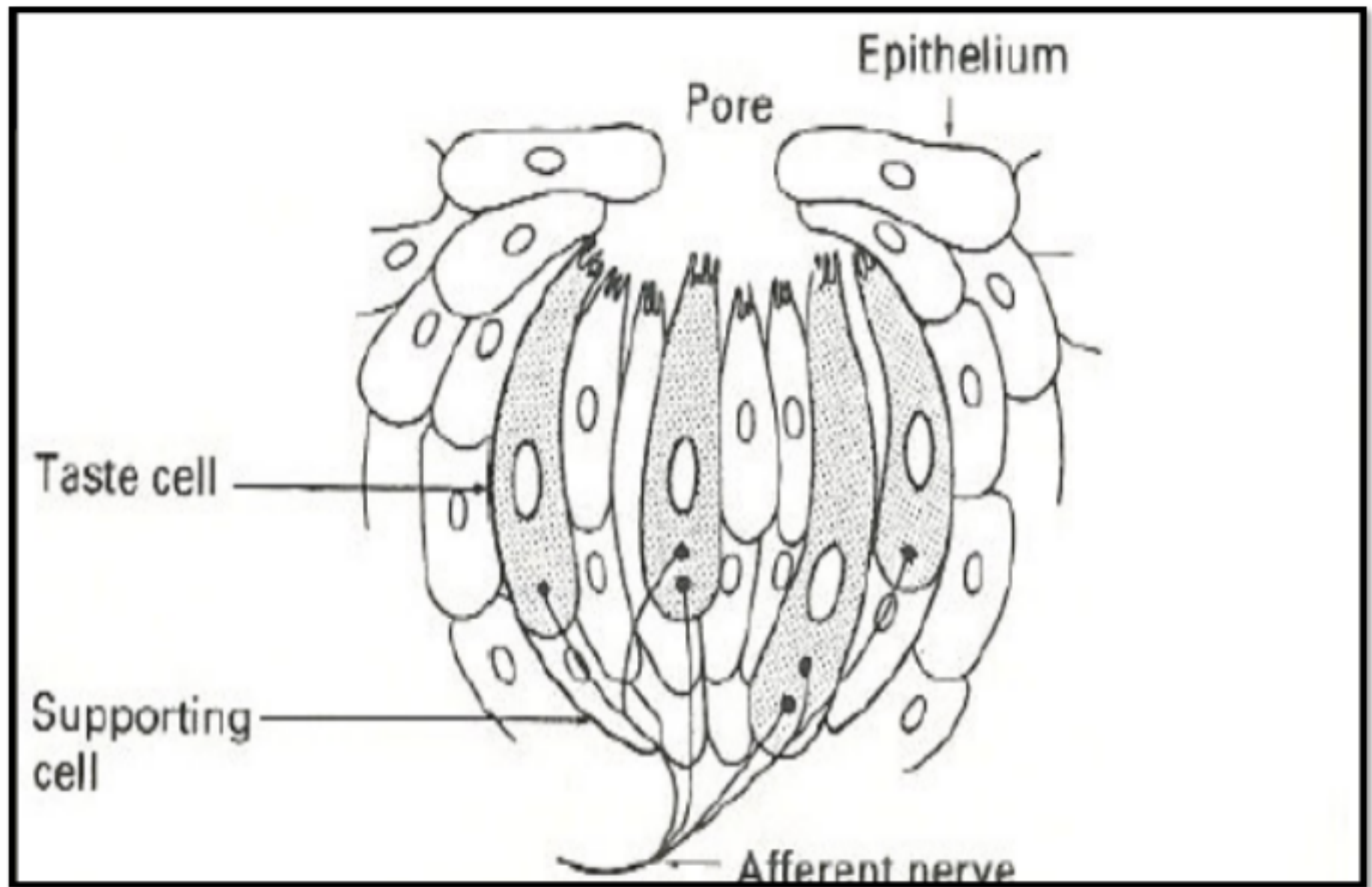
Taste and Smell

- **Receptors** : Chemo receptors
- **Importance:**
- **Quality control:**
 - Smell and Taste provide check point for substances available for ingestion
 - (i.e. smell and taste receptors induce a **pleasurable sensation** (good- tasting and pleasurable food to seek) or **objectionable sensation** (bad-tasting ,a potentially toxic substance to avoid)
 - Smell and Taste are less sensitive and less important in **affecting our behavior** than in animals.

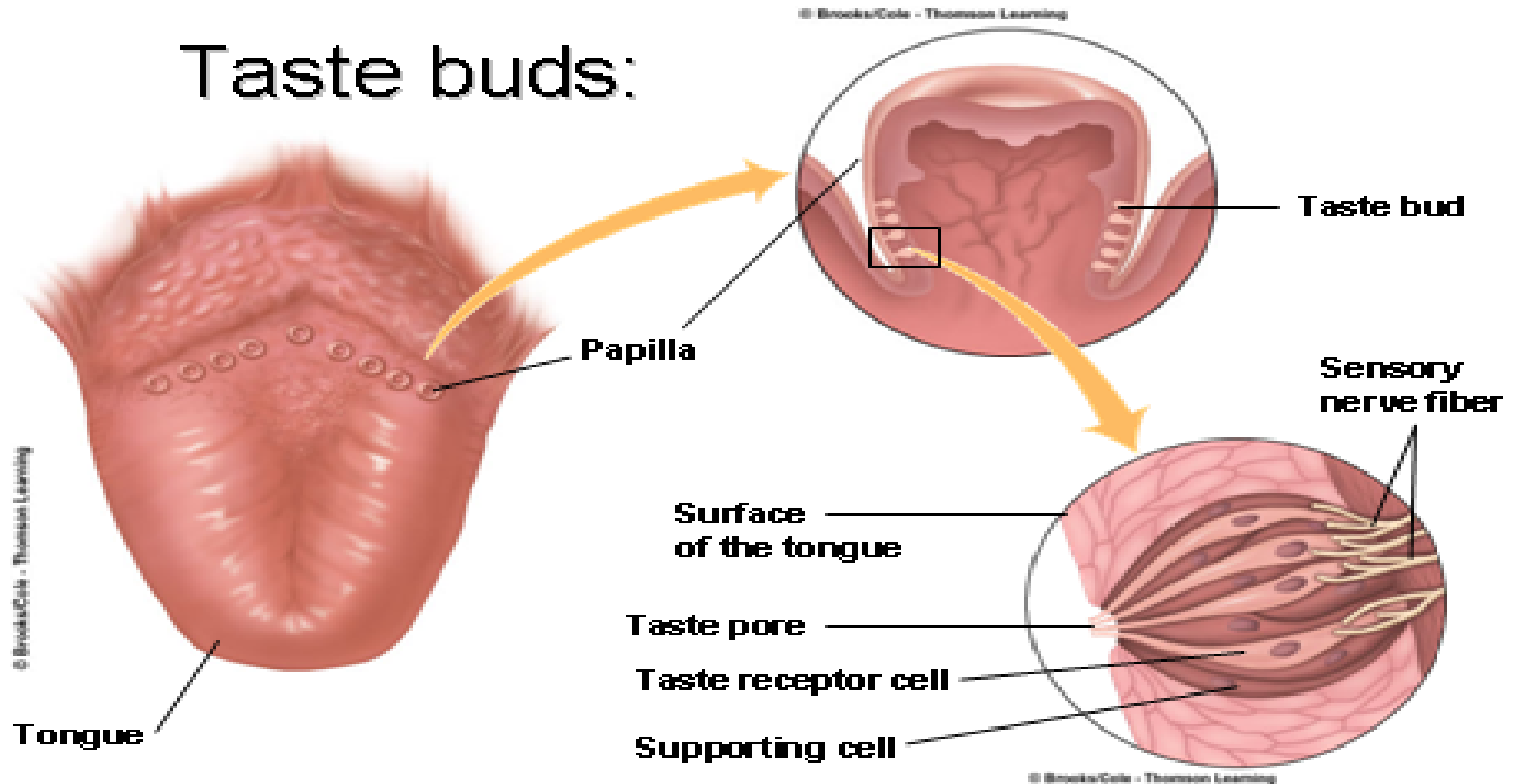


Taste Sensation

- **Receptors:** Chemo receptors are backed in taste buds
- **Taste buds** (about 10000, are found in the oral cavity and throat) ,with greatest percentage on the upper surface of the tongue
- A. **Structure:** About 50 spindle shaped receptor cells packaged with supporting cells.
- B. **Taste pore:** Small opening in each taste bud through which fluids in the mouth come into contact with the surface of its receptor cells



Taste buds:



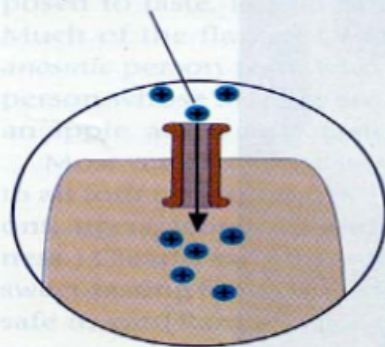
- **Taste receptor cells:**
- Have many folds or microvilli that increases the surface area of contact with oral contents
- **Receptors :** located in the plasma membrane of the microvilli that bind selectively to only chemicals in solution (solids or liquids that have been dissolved in saliva)
- Receptors have a life span of about 10 days and are replaceable

The mechanisms of stimulation of specific taste receptors

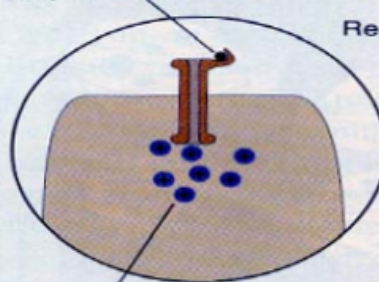
- **Sour** is detected when acids containing H^+ block K^+ channels.
- **Salt** probably acts by depolarization of salt receptors due to influx of Na^+ .
- **Bitter** stimuli act by a G protein (G_o), which activates phospholipase C, leading to an increase of intracellular inositol triphosphate (IP_3), leading to Ca^{2+} release from the endoplasmic reticulum.
- **Sweet** stimuli work through a G protein (G_s), which activates adenylate cyclase, thus increasing cAMP, which leads to decreased K^+ conductance.



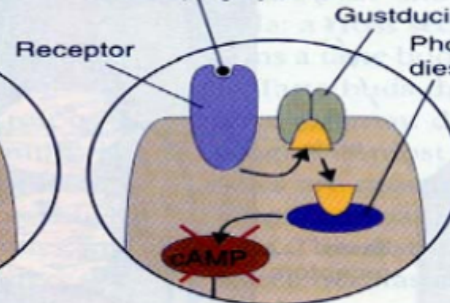
Sodium ions enter ion channel, depolarize membrane



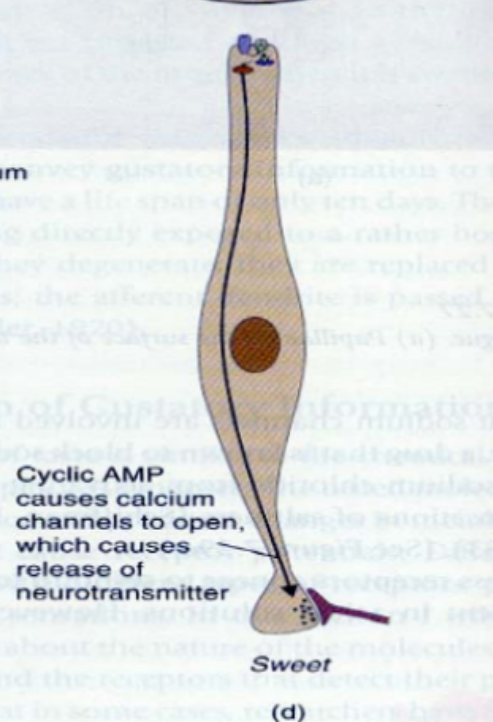
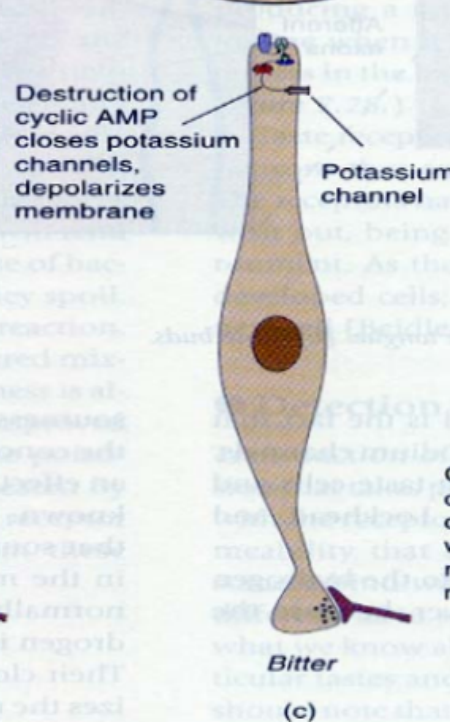
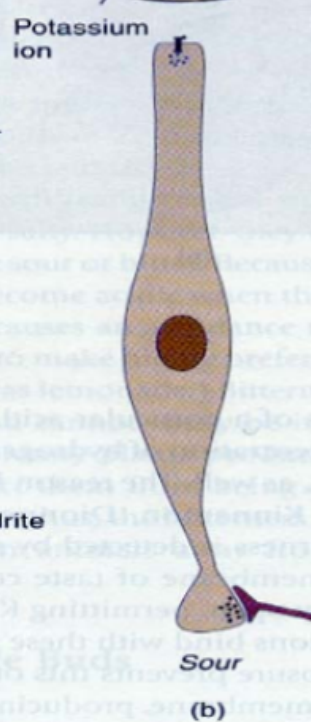
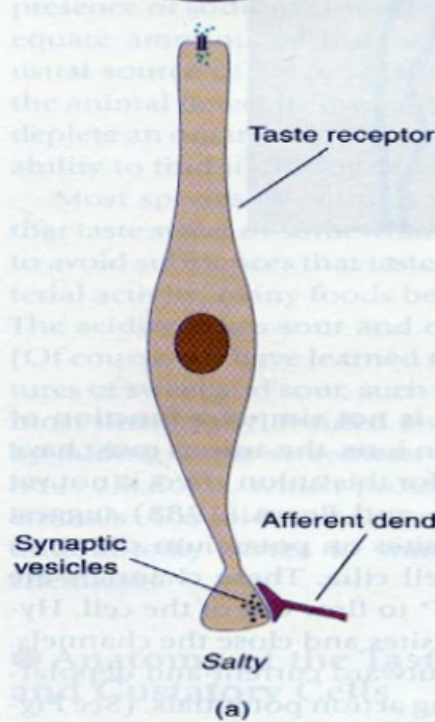
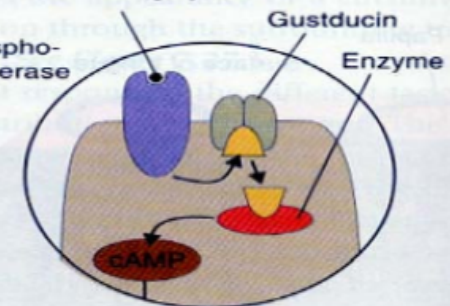
Hydrogen ion binds with receptor, closes potassium channel. Potassium ions cannot leave cell, membrane depolarizes.



Bitter substance binds with receptor, activates gustducin, which activates phosphodiesterase, destroying cyclic AMP

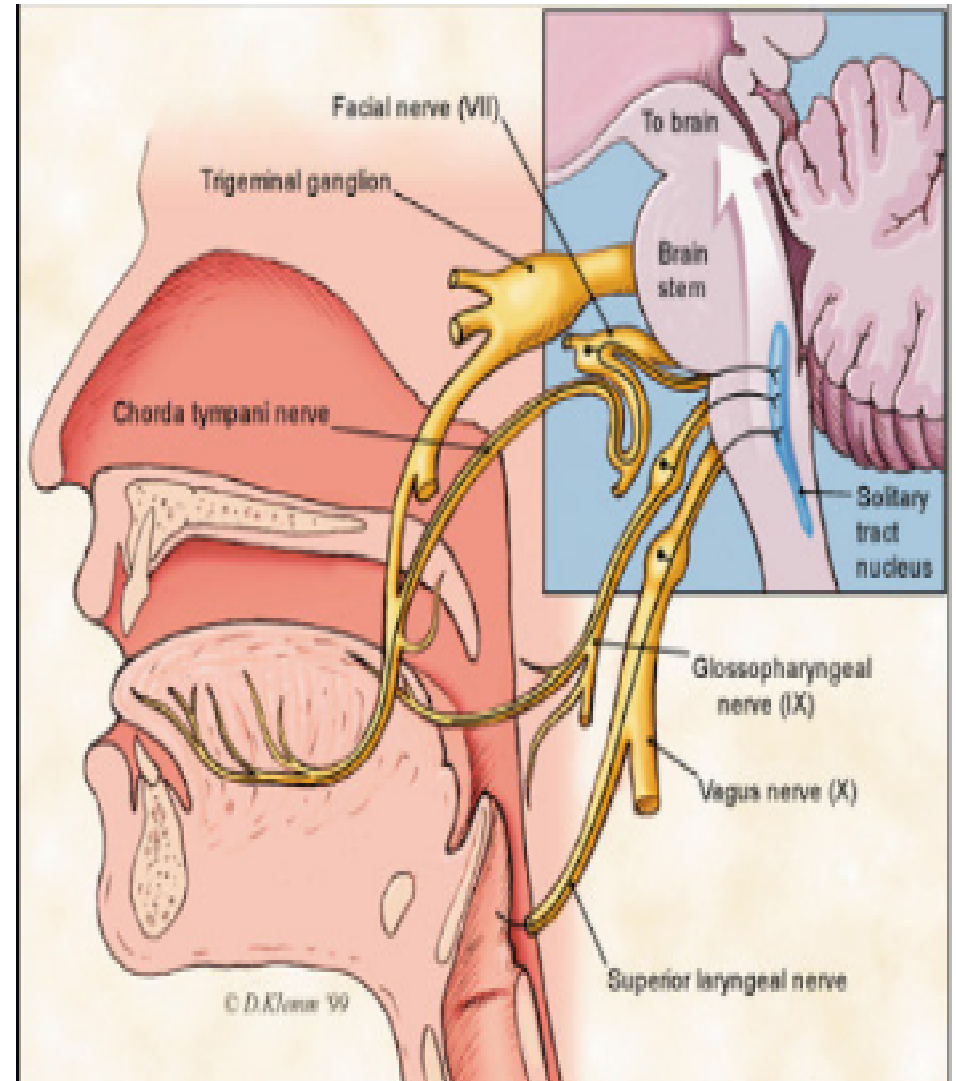
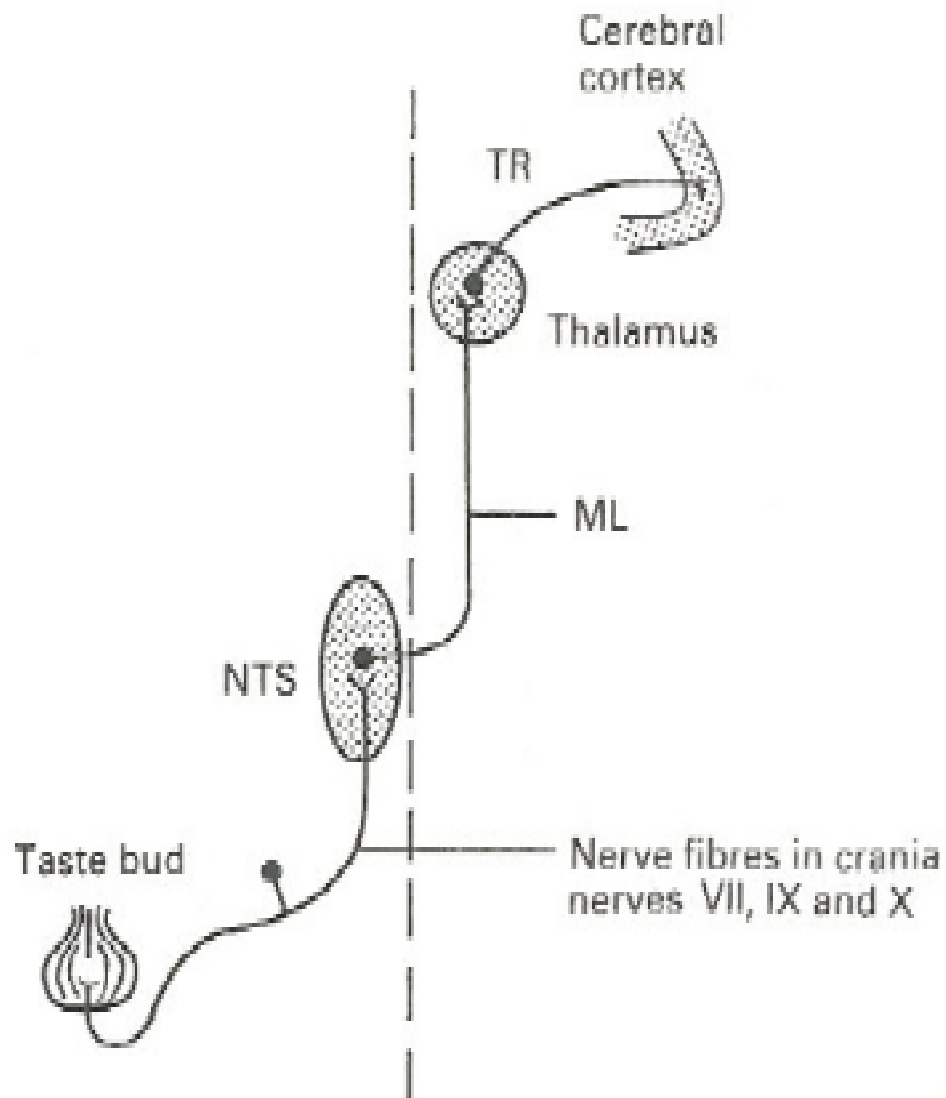


Sweet substance binds with receptor, activates gustducin, which activates an enzyme, producing cyclic AMP



Pathway of taste

1. Receptor potential summate and generate action potential within nerve fibers terminals that synapse with receptor cell
2. Several cranial nerves carry action potentials (sensory signals) from various taste buds in different parts of the mouth and tongue
3. Synapse occur at the brain stem then Thalamus to end in the parietal lobe at (Cortical Gustatory area), area just adjacent to the tongue area of the somatosensory cortex
4. The brain stem projects signals to the hypothalamus and limbic system for quality control and behavioral aspects associated with taste sensation



Taste Categories (Modalities)

- **4 primary tastes:** Salty, Sour, Sweet and Bitter
- Actually we can distinguish thousands of tastes ,
- this is due to the fact that each taste receptor cell responds to a varying degree to **all taste types** but is generally more responsive to **one taste** modality (type).
- the taste produced depends on the various degrees of stimulation of taste buds (this very similar to color vision in the eye where stimulation of the 3 primary cones types by various degrees give rise to different colors)

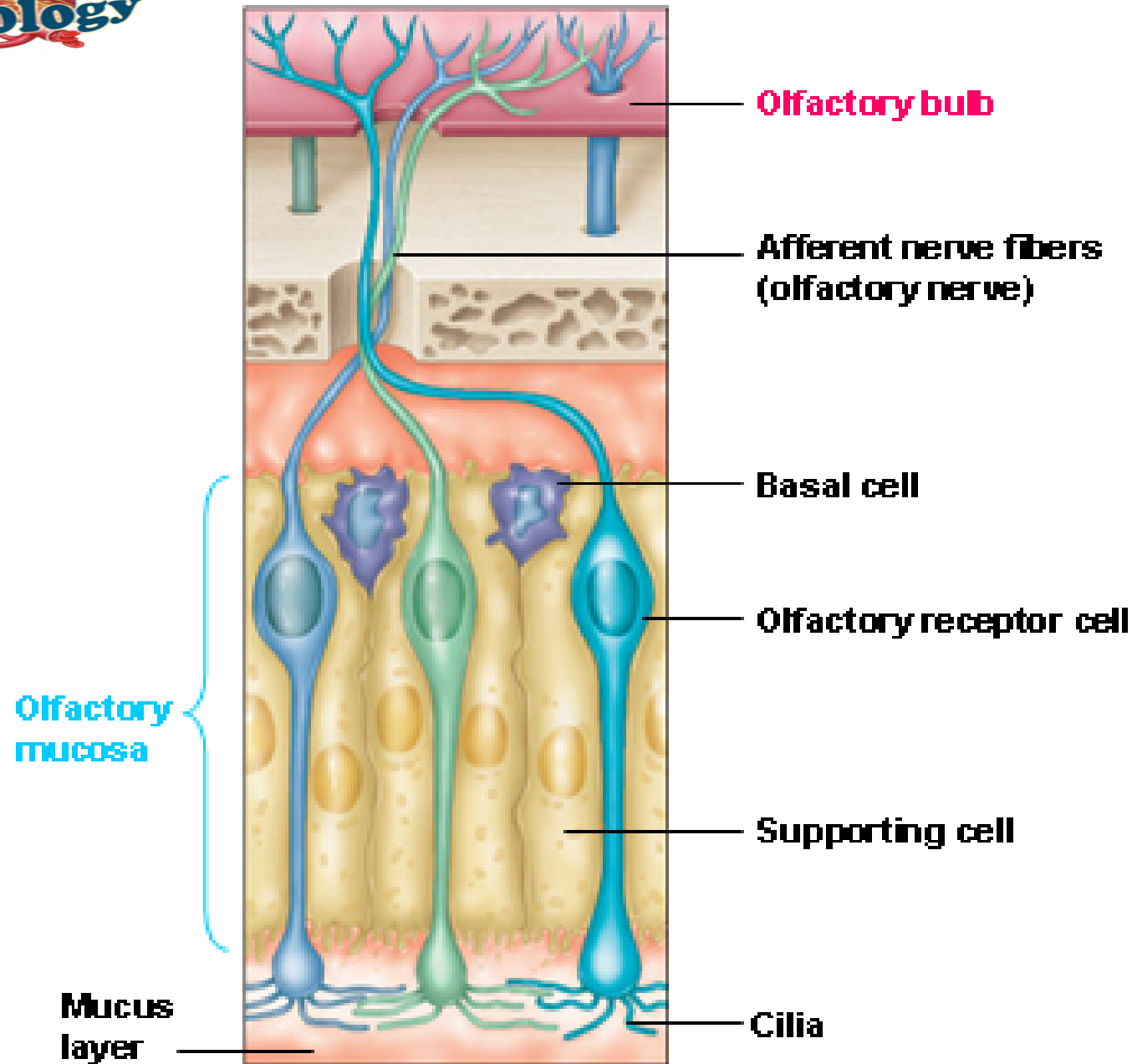
- **Taste and odor:**
- Temporary loss of sense of smell as during nasal congestion (due to cold) reduces markedly taste sensation although taste receptors are unaffected by the cold
- Other factors affect taste include temperature, texture of food as well as past experience of food (psychological factors)

4-bitter taste	3-Sweet taste	2-Sour taste	1-Salt taste	
-Alkaloids (caffeine, nicotine, strychnine, morphine) - Poisonous substances	Glucose and other sweeteners (saccharin, aspartame)	Acids (with free H^+) (e.g. citric acid in lemons)	Chemical salts (NaCl =table salt)	Stimulus



Smell (Olfaction)

- **Olfactory mucosa**
- **Site:** ceiling of the nasal cavity
- **Cells:** 3 types: **1.** olfactory receptors **2.** supporting cells (secrete mucous) **3.** basal cells (precursors for new olfactory cells)
- **Olfactory receptors:** about 5 millions receptors of 1000 different types, specialized endings of afferent neurons and not separate cells
- They are replaced every 2 months, the entire neuron including its afferent axon projecting into the brain is completely replaced, they are the only neurons capable of cell division

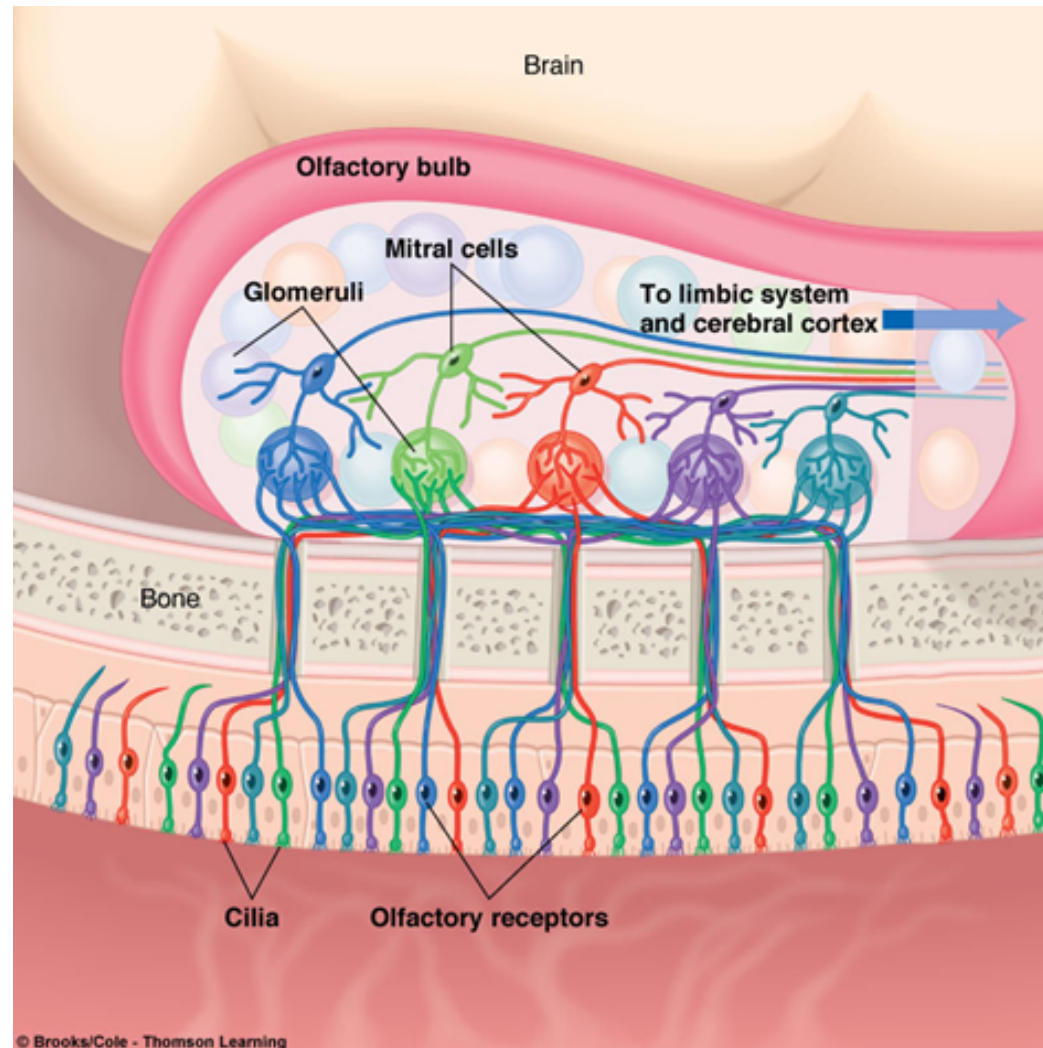


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- During **smell detection** each receptor **respond to one component** of the odorant molecule (dissected) and not the whole molecule, so different parts of an odor are detected by different receptors
- **Olfactory nerve** :
- Formed by the axons of receptor cells

- The receptor portion of the olfactory receptor cells is formed of enlarged knob containing several long cilia extend to the surface of the mucosa
- Odorant molecules attach to binding sites present in the cilia
- During normal breathing: Odorants reach the receptors only by diffusion
- During eating odorants can reach the olfactory mucosa by moving up to the nose from the mouth through the pharynx

Olfactory receptors and Olfactory bulb



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✓ Characters of substance to be smelled:

1. Sufficiently volatile (easily vaporized)
2. Sufficiently water soluble so can dissolve in the mucous layer